

average value of 1.4925, which is accurate. But Option B is less precise than Option A.

3 Ans: C

$$\begin{aligned}\text{After 3 s, } x_2 - x_1 &= (u_1t + \tfrac{1}{2} a_1t^2) - (u_2t + \tfrac{1}{2} a_2t^2) \\ 36 &= (0 + \tfrac{1}{2} a_1(3^2)) - (0 + \tfrac{1}{2} a_2(3^2)) \\ a_1 - a_2 &= 8\end{aligned}$$

$$\begin{aligned}\text{After 5 s, } x_2' - x_1' &= \tfrac{1}{2} a_1(5^2) - \tfrac{1}{2} a_2(5^2) \\ &= \tfrac{1}{2} (5^2)(a_1 - a_2) \\ &= \tfrac{1}{2} (5^2)(8) \\ &= 100 \text{ m}\end{aligned}$$

4 Ans: B